

How the lottery works

Three rounds.
One published method.

Fairness without Stress

The algorithm gives everybody a chance at one club before anybody gets a second, and everybody a chance at a second before anybody gets a third. Within each round, a seeded lottery decides who is considered first.

What fairness means here

1 Equal deadline Every valid, on-time entry joins the same lottery. Earlier submission has no advantage.	2 Spread opportunity One place each is considered before second and third places.	3 Respect choices A child gets only a club they requested, tried in their own preference order.	4 Make it auditable A recorded seed and version make the result reproducible.
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Inputs to the lottery

Each child Opaque ID + 1-3 ranked choices	+	Each club Name + available capacity
Run seed Random value, saved for audit	+	Algorithm version Published rule set

The timestamp is retained in result files for administration but deliberately excluded from every priority calculation.

A repeatable random order

For each eligible child in each round, Clubstonbury calculates a SHA-256 digest from:

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algorithm version : seed : round : round number :
child : opaque child ID
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Sorting these digests creates the round's lottery order. A different round number creates a fresh order. The child ID breaks the extraordinarily unlikely case of equal digests.

Not included: name, timestamp, CSV row, number of choices, existing choice rank or personal characteristics.

The three allocation rounds

First place 1 - Every child is eligible. - Use round 1 lottery order. - Try their requested clubs in order of preference until finding one that is open. - Stop after at most one allocation.	Second place 2 - Only children with exactly one place are eligible. - Create a separate round 2 order. - Try remaining requested clubs in preference order. - Stop after at most one more.	Third place 3 - Only children with exactly two places are eligible. - Create a separate round 3 order. - Try the last unallocated requested club. - Stop after at most one more.
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At every step: a place is allocated only when capacity remains. A club can never exceed its entered capacity, and a child can never receive an unrequested club.

Four children, four places

Seeded order:
B, C, A, D

Suppose Football has 1 place, Drama has 2 and Coding has 1. Round 1's hash order is B, C, A, D:

Turn	Child's choices	What the algorithm does	Result
B	Football, Coding	Football is open.	Football
C	Drama, Football	Drama is open.	Drama
A	Football, Drama	Football is full; next-ranked Drama is open.	Drama
D	Coding, Drama	Coding is open.	Coding

Everyone has one club, and every capacity is now full. No one can receive another place in rounds 2 or 3. A's lower-ranked Drama choice helped A; ranking it did not change A's chance at Football.

Waiting lists

After all three rounds, each unsuccessful request becomes an entry on that club's waiting list. Each club gets its own seeded SHA-256 order.



- A successfully allocated request is excluded.
- Original choice rank and current allocation count appear for context, but do not change waitlist position.
- If a place is returned, use the next eligible child without reordering.

Then mop-up

The deterministic waiting list protects the original applicants. Only after a club's original list is exhausted should leftover places enter mop-up.

Mop-up is first come, first served. Offer residual places to late applicants or children making new choices in the order those requests arrive. Do not rerun the original lottery to add them.

Properties you can check

- **Preference-order safe:** a lower-ranked choice is considered only after higher-ranked choices are unavailable at that turn.
- **No choice-count bonus:** listing fewer clubs does not improve priority.
- **No timestamp race:** all on-time applications are treated equally.
- **Capacity-safe:** remaining capacity is checked before every allocation.
- **Requested clubs only:** no automatic substitute is invented.
- **Reproducible:** the same inputs, seed and version give the same results.

Reproduce and audit a run

1 Prepared applications CSV	2 Confirmed club capacities	3 Exact run seed	4 Algorithm version
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Keep the downloaded result ZIP. Its manifest contains the seed, capacities, algorithm and product versions, run time, warning record and SHA-256 hash of the source CSV. The timestamp documents when the button was pressed; it does not affect the allocation.

Scope: Clubstonbury applies this published model; it does not decide eligibility or legal priority. The administrator must confirm the model suits the school's policy, check all data and results, and apply any duties or reasonable adjustments outside the algorithm.